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Open-plan office noise: Cognitive performance and restoration

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ABSTRACT

The aim of the present study was to investigate cognitive, emotional, and physiological effects of two open-plan office noise conditions (high noise: 51 LAeq and low noise: 39 LAeq) during work in a simulated open-plan office, followed by four restoration conditions (river movie with sound, only river sound, silence, and office noise) after the work period. Students (N = 47) went through one practice session and two experimental sessions, one each with the low and high noise conditions. In each experimental session they worked for 2 h with tasks involving basic working memory processes. We also took physiological measures of stress (cortisol and catecholamines) and self-reports of mood and fatigue. Analyses indicate that the participants remembered fewer words, rated themselves as more tired, and were less motivated with work in noise compared to low noise. In the restoration phase the participants who saw a nature movie (including river sounds) rated themselves as having more energy after the restoration period in comparison with both the participants who listened to noise and river sounds. Remaining in office noise during the restoration phase also affected motivation more negatively than listening to river sounds or watching the nature movie. The findings bear on the appropriateness of open-plan office designs and the possibilities for restoration available in office settings.

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1. Introduction

Research has often reported lower satisfaction, lower work productivity, and poorer health among the employees working in open-plan offices compared to traditional offices (see reviews by Navai & Veitch, 2003; Oommen, Knowles, & Zhao, 2008; Rashid & Zimring, 2008). Research has also indicated that the majority of the negative effects seem to be associated with an increase in background noise and ensuing distraction (Hedge, 1982). The results are, however, mainly from studies in field settings with self ratings of different work conditions rather than from laboratory studies with controlled conditions and objective measures of cognitive performance and acute stress effects (see review by De Croon, Sluiter, Kuijer, & Frings-Dresen, 2005). Such studies can also aid the understanding of different sub-processes within working memory activated with office work. Such studies can also address relatively little studied issues concerning methods for recovery from work in office settings.

In the present experimental study we approached two main issues concerning open-plan offices. First we examined the effects of noise on basic working memory processes, self-ratings (mood and fatigue) and physiological measures of stress (saliva and urine samples). Second, we examined the effects of varying sound conditions on restoration at work.

1.1. Office noise effects on working memory and stress

Working memory (WM) processes are of crucial importance when working with complex tasks because they process information necessary for the task at hand and temporarily store and handle the needed information. WM has a limited capacity (Baddeley, 2002) and performance can decline if a task is performed in competing background noise (Banbury, Macken, Tremblay, & Jones, 2001; Jones, 1990; Sörqvist, Halin, & Hygge, 2009).

Noise appears to affect different WM tasks in different ways. For example, noise (such as irrelevant speech) impairs performance in proofreading (e.g., Smith-Jackson & Klein, 2009; Venetjoki, Kaarlela-Tuomaala, Keskinen, & Hongisto, 2006), serial recall (for a review, see Jones & Morris, 1992), mental arithmetic (e.g., Banbury & Berry, 1998; Schlittmeier, Hellbrück, Thaden, & Vorländer, 2008), reading comprehension (REFS), operation span, and tasks activating prior knowledge from long-term memory (Haka et al., 2009). Even if the majority of these tasks involved memory and seriation at some point (encoding and/or rehearsal), the studies particularly

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interested in the effects of open office noise have mainly taken a macro perspective on working memory. According to Roper and Juneja's (2008) categorization, only three of the 59 studies they reviewed included tasks which measure the subsystems of working memory and only five studies included tasks which measure the earlier unitary model of short-term memory. Thus, there is a need for further applied noise studies which make use of the different measures recently developed in working memory research.

One way to approach and investigate how WM is affected by noise applies the well known multicomponent model of working memory (Baddeley & Hitch, 1974; further developed by Baddeley, 2000). According to this model there are two modality-specific stores handling verbal or visual stimuli (i.e., the articulatory loop and the visuo-spatial sketchpad, respectively), together with one episodic buffer capable of multidimensional coding and one executive function that controls the other subsystems.

Another approach does not include modality-specific buffers such as the articulatory loop. Instead the focus is on the limited capacity of working memory and how a set of integrated processes can get disturbed by noise (Just & Carpenter, 1992). In the present study working memory will mainly be seen from different measures that have been developed to tap the integrated processes involved in complex tasks, rather than through further investigation of the hypotheses of working memory.

Earlier studies have indicated that task difficulty is important for the degree of disturbance by noise (e.g. Kjellberg & Sköldström, 1991), and depending on which task is performed, there seem to be different executive functions involved. Results from Miyake et al. (2000) suggest that updating (the ability to update information in WM), shifting (the ability to shift between mental sets) and inhibition (the ability to inhibit responses) are moderately correlated but clearly separable functions, and that it is possible to distinguish them according to their relative contribution to performance in complex tasks. We therefore employed tasks developed to tap these three basic functions, to explore whether these functions are differentially affected by office noise.

Further research (e.g., Friedman & Miyake, 2004) has also suggested that there are at least two inhibition mechanisms that must be distinguished. The first mechanism is *response inhibition*, which inhibits motor responses or distracting stimuli and can be tapped by tasks such as the Stroop task (e.g., Venetjoki et al., 2006), the Flanker task (e.g., Heitz & Engle, 2007) and the Sustained Attention to Response Test (SART; e.g., Manly, Robertson, Galloway, & Hawkins, 1999). The second mechanism is *cognitive inhibition*, which inhibits no-longer relevant information that was once activated in working memory and can be tapped by tasks that involve proactive interference (PI; e.g. Lustig, May, & Hasher, 2001). Therefore, tasks related to each of these mechanisms (i.e., SART and PI) were included in the present investigation of fatigue- and office noise effects.

Only a small number of studies have reported objective physiological assessments of the acute stress effects of office noise. These studies have mainly used measures of stress hormones, heart rate variability, blood pressure and activity in the trapezius muscles (e.g., Evans & Johnson, 2000; Kristiansen et al, 2008; Sloan, 1991; Waye et al., 2002). We were also interested in finding out whether our acute office noise exposure would show up in physiological measures, and whether physiological reactions mediate effects on cognitive performance during exposure to office noise. Hence, we include urine- and saliva samples to measure the changes in stress hormone production.

1.2. Restoration

If stress increases and performance declines while working in office noise, the question arises whether these changes can be moderated if a short restoration period is included, and whether the ensuing changes vary with the content of the restorative period.

Studies of restoration have mainly shown that nature environments are restorative to be in or to look at when cognitively fatigued (e.g., Berto, 2005; Hartig, Böök, Garvill, Olsson, & Gärling, 1996). These studies focused on the *visual* perception of the environment and not on the *sound* conditions per se. There are, however, some results indicating faster and more complete restoration (according to both performance and physiological measures) for participants exposed to nature movies with environmental sounds, in comparison to those exposed to other environments including sounds (Laumann, Gärling, & Stormark, 2001, 2003; Ulrich et al., 1991). Some studies have also shown a restorative advantage of natural versus urban settings in field conditions that included the sounds typical of each type of environment (Berman, Jonides, & Kaplan, 2008; Hartig, Evans, Jamner, Davis, & Gärling, 2003; Hartig, Mang, & Evans, 1991).

Khalfa, Bella, Roy, Peretz, and Lupien (2003) have also shown the restorative effects of positive sounds only. They compared recovery from sitting in silence with listening to soft music, and found that music promoted faster decline in salivary cortisol, measured after 15 min into the relaxation period. However, no studies (to the authors' knowledge) have investigated whether positive nature sounds by themselves can promote restoration and ensuing cognitive performance.

1.3. Aims and hypotheses

The aims of the present study thus were to further specify how noise in open-plan offices affects cognitive performance and acute stress, and to test whether it is possible to promote cognitive restoration with exposure to pleasant sounds and film-clips of pleasant nature environments after having been exposed to aversive sounds. Our hypotheses are as follows:

Hypothesis 1. Participants' *cognitive performance* will decrease more in high noise compared to low noise.

Hypothesis 2. Participants will have a larger increase of *stress hormones* from before to after the work session in high noise compared to low noise.

Hypothesis 3. Participants' *self ratings* of tiredness and motivation will decrease more with work in high noise compared to low noise.

Hypothesis 4. Participants will *restore* from fatigue (measured as performance on cognitive tasks, self ratings and cortisol levels) to differing degrees as a function of environmental conditions, including sound variations. More precisely, we expect the following order of restorativeness: a nature movie with nature sounds (positive stimuli for two senses), just nature sounds (positive stimuli for one sense), silence (no stimuli) and noise (negative stimuli).

2. Method

2.1. Design

To test these hypotheses we designed an experiment with two acoustic environments (low noise, high noise) varied within subjects and four restorative conditions (nature movie, nature sound, silence, noise) manipulated between subjects.

2.2. Participants

The participants were 47 persons with normal hearing and vision recruited from the University of Gävle (27 female; mean

age = 26). Participants were informed of the study's nature and risks before participating and they were randomly assigned to conditions. Participation in all sessions of the experiment was compensated with 990 Swedish crowns.

2.3. Research setting

The research was carried out in an office laboratory (63 m^2) at the University of Gävle (Fig. 1). The room was designed to simulate a neutral open-plan office including windows to a white room with simulated outdoor lighting.

The workstations were separated with 1.43 m high screens. Each cubicle was 1.24 m wide. At each end of the set of cubicles a pane of glass was installed to better avoid direct sounds from the loudspeakers to the participants sitting close by. Behind the participants on each side were also 3.60 m wide × 1.80 m high screens to better avoid direct sounds from the loudspeakers behind.

Ambient conditions in the indoor environment were fairly constant throughout the experiments. The room temperature at the work place varied between 20.7 and 21.6 °C during the experimental sessions. The airflow was 20 l/s for each person in the experiment, and the CO_2 concentration was always well below 1000 ppm. Luminance was set to 480–520 lx incident on the height of the seating in each work station and met the current recommendations in Sweden for work with computers (SS-EN 12464-1, 2003). The luminance in the room outside the window was set to 1950 lx measured at the middle of the window.

2.4. Noise conditions

The noise used in the experiment was recorded in four positions in an actual open-plan office in Sweden. From the multi-channel recording, 1 h of office noise was extracted and edited and reproduced in the test room with eight loudspeakers, two at each wall, and one subwoofer. Two noise conditions were designed, one with a high- and one with a low noise level. In the high noise condition, additional phone and speech signals were added to the office noise. The phone signals consisted of different mobile tunes and stationary telephones ringing. The speech signals were conversations recorded from radio, from which we cut away one of the voices to simulate telephone conversations. The noise was reproduced with an equivalent A-weighted sound pressure level (L_{Aeq}) of 51 dBA in the room, since a common noise level for open-plan offices is around 50 dBA (Venetjoki et al., 2006). In the low noise condition, the office noise, including the additional phone and speech signals, was reduced L_{Aeq} in the room by 12–39 dBA. The filter was used to check if noise levels reduced by a wall screen would give positive effects in open-plan offices. The 1 h office noise tape was looped once, and the sound signals in the latter hour were rotated 180° to equalize noise exposure throughout the room. The phone and speech signals in the high noise condition were added independently to each 1 h sequence but were balanced in terms of level, duration and frequency.

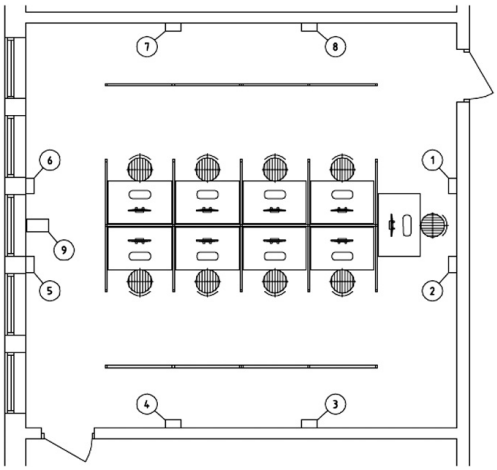


Fig. 1. The simulated open-plan office contained eight workstations. The noise was played back from 8 loudspeakers (1–8) and one subwoofer (9). The experimental leader sat behind the partition at one short side of the block of work stations. On the opposite side were windows to a white room with luminance corresponding to an outdoor environment.

On the basis of A-weighted sound pressure level measurements carried out in the test room, equal sound conditions in the eight workstations were confirmed. The measurements also showed the same L_{Aeq} in the first and second hour of the reproduced office noise. Statistical analysis of the recordings showed that high and low noise conditions were similar in respect with time variation, spectra, and speech-to-noise ratio. The percentile analysis showed that the range of sound pressure levels was similar in both noise conditions (see Table 1). That is, statistical distribution of sound pressure levels was similar except that the equivalent levels differed by 12 dB. The Speech Transmission Index (STI) was almost the same in both noise conditions.

2.5. Restoration conditions

After 2 h of work in office noise the participants went through a restoration period for 7 min. There were four different restorative conditions: watching film clips of rivers while listening to corresponding river sounds, listening to the river sounds without the film clips, sitting in silence, or continuing to listen to the office noise from the high noise condition. The nature film contains changing soothing scenes of streaming water and was filmed in different landscapes of the USA (used with permission from Isis Visuals, Ede, The Netherlands; www.natursounds.ca). The river sounds were recorded with a binaural technique to give a better simulation of being in the real sound environment. The river sound is shallow and slow with a birds chirping in the background at a few points. The different film and sound clips were presented to the participants via the computer at their work station.

2.6. Measures

2.6.1. Cognitive performance

We used several different cognitive tasks to measure the effects of the noise and restoration manipulations. Both a cognitive inhibition task and a response inhibition task (further described below) were included to measure fatigue/restoration and were performed three times: in the beginning (pretest), after 1.5 h of work (postwork), and after the restoration period at the end of the experimental sessions (postrest). The other cognitive tasks were included

Table 1
The analysis of two noise conditions in respect with STI and time variation.

		High noise (51 dBA)	Low noise (39 dBA)
Range of STI values		0.00 to 0.66	0.00 to 0.58
Overall variation of SPL, $L_{A,eq,1s}$	dB	40 … 60	31 … 50

only once to measure the effects of the different noise conditions (see Fig. 2 for an overview of the task order). The same order of measures was used for all participants to control the timing between task and noise, since we played back the noise from loudspeakers (and not by headphones for each individual). It would also have been extremely difficult to run this complex experiment with random orders of measures. Since we were not trying to make an evaluation of each specific test, we were ready to take the confounding between tests and time on task.

2.6.1.1. Proactive interference. Proactive interference *(PI)* was measured with a task which taps the ability to inhibit information that once was relevant for recall but has since become irrelevant for the task (Peterson & Peterson, 1959). The participants learned words from ten lists with five words each. The lists were composed of words drawn from the same category. After a list with five words they had to recall the words from the most recent list and ignore (suppress) words from earlier lists. We used the following categories of words: "fruits and vegetables", "animals", "cloths and accessories", "kitchen- and gardening tools", "countries" and "cities in Sweden". The words were presented on the computer screen every 1000 ms and remained for 1000 ms. The orders of the PI-categories within each experimental session were counterbalanced. The following four scores were considered: (1) PI-points: the sum of words recalled from the correct list; (2) PI-pro: the sum of error words (i.e. from earlier lists); (3) PI-no answers: the sum of words not remembered; and (4) PI other errors: the sum of error words (i.e. new words not coming from earlier lists).

2.6.1.2. Response inhibition. Response inhibition was measured with the Sustained Attention to Response Test *(SART)*, which taps the ability to inhibit a physical response and/or sustain attention (Manly et al., 1999). Digits from one to nine were presented repetitively and the participants were told to respond with a key press to all numbers except number three. A digit was presented on the computer screen once every 1000 ms and remained for 1000 ms. Each trial consisted of 100 digits, where 22% were targets (i.e., the number 3) to which the response was to be inhibited. The following scores were considered: (1) the sum of errors of commission (i.e., the number of times the response to the number three was not successfully inhibited); (2) the sum of errors of omission (i.e., the number of times a response was not made to a number which should have been responded to), and; (3) reaction time (i.e., the mean response time in milliseconds for all numbers except number three).

2.6.1.3. Flanker task. We adopted a version of the flanker task (Eriksen & Eriksen, 1974; Heitz & Engle, 2007) as another computerized measure of response inhibition. The participants viewed strings of five letters composed of H and S (e.g. HHHHH, HHSHH). They were told to respond to the central letter by pressing "h" or "s" on the keyboard, inhibiting the other letters (flankers). Before the first letter string was presented in each trial, the symbol "##" was presented and indicated where on the screen the strings were going to be shown. The orienting symbol and the letter strings remained visible for 500 ms. One block consisted of 47 strings, and the mean reaction time for one block was measured.

2.6.1.4. Logical problems. In this task, the participants were presented with logical problems and were told to answer whether the conclusion drawn was true or false (e.g., All A are B. Some B is C. Conclusion: some A is C. Is this conclusion true?). They were told to respond as quickly and correctly as possible. The sum of correct answers was considered.

2.6.1.5. Operation span and reading span. In the operation span test, the participants were presented with sets consisting of a mathematical operation and a word (e.g., Is $(4/2) + 1 = 3$? and then the word "Rose"). The participants were told to respond "yes" or "no" if the mathematical expression was true or not by pressing "Ä" on the keyboard for yes and pressing "A" for no. They were further told to remember the word presented after each mathematical operation. After three to seven tasks they were asked to recall all the words presented. Doing so requires the immediate suppression of the mathematical operations and suppression of words from earlier lists (see Turner & Engle, 1989). The scores were the sum of correct answers in the mathematical operations and the sum of correct words remembered.

The purpose of the reading span task was similar to that of the operation span task. The participants were to decide if a sentence

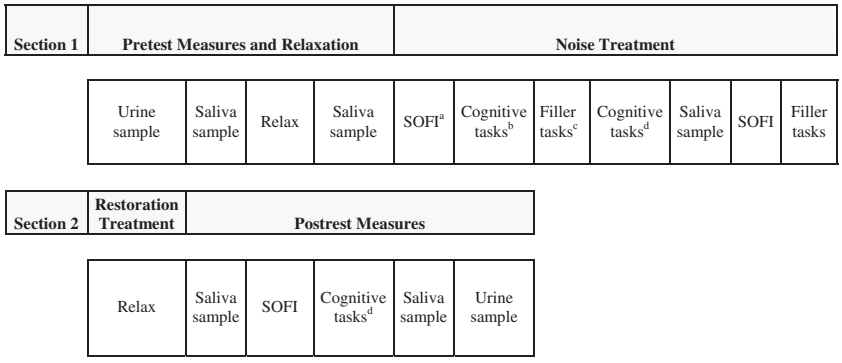


Fig. 2. Overview of the procedure for one experimental session. Section 1 focused on the effects of noise (low and high noise). It was followed immediately by Section 2, which focused on the effects of different options for restoration (Movie of a flowing river with sound, River sound only, Silence, and Noise). [a]Swedish Occupational Fatigue Inventory. [b]The order of the cognitive tasks was as follows: Sustained Attention to Response Test (SART), Proactive interference (PI), Flanker, Reading span, Serial recall – letters, Updating, Operation span, Logical problems, Reading comprehension, Serial recall – numbers, Shifting. It took circa 110 min to complete the tasks. [c]The filler tasks were included to make all participants begin the next phase at the same time. [d] The order of the cognitive tasks at *Postwork* and *Postrest* was as follows: Sustained Attention to Response Test (SART), Proactive interference (PI). It took about 10 min to complete these tasks.

was normal or abnormal and then remember a two-digit number (e.g., "The cloud laugh yellow", Is the sentence normal? and then the number 32). After three to seven tasks they were to recall all the recent numbers presented (see Daneman & Carpenter, 1980).

2.6.1.6. Reading comprehension. This task consists of ten short texts presented sequentially. The texts were adopted from an earlier experiment by Sörqvist et al. (2009). The participants were required to answer one question per text with four alternative answers (only one is correct). The participants had 2 min to answer each question by pressing a button representing their answer (a, b, c, d). If they failed to give an answer within the time limit, the next text was presented. Half of the questions were about the meaning of the text and half of the questions were about a missing word in the text. The score was the sum of correct answers.

2.6.1.7. Serial recall (numbers and letters). In one version of this task, the participants were told to remember a string of seven one-digit numbers and then recall them in the correct order. Each number was presented for 1000 ms and then it was a pause of 600 ms before the next number was presented. Each block consisted of 10 lists and only numbers written at their correct positions were scored as correct. In the other version of this task, they were told to remember a string of seven letters presented in the same way.

2.6.1.8. Shifting. Shifting measures the ability to shift between mental sets. We adopted the number-letter task from Rogers and Monsell (1995), which requires switching between two different categorization operations. The task consists of number-letter pairs (e.g., 3E) presented in a clockwise rotation around all four quadrants on the computer screen. When the number-letter pair is presented in the top two quadrants, the participants are to indicate whether the number is even or odd, and when presented in the bottom two quadrants, they are to indicate whether the letter is a vowel or a consonant. The participants respond by pressing different keys for even/odd and vowel/consonant answers. A number-letter pair is presented on the computer screen every 1000 ms and remains for 1000 ms. Before the participants begin the real task they have three practice trials: (1) to respond to number-letter pairs in the top two quadrants; (2) to respond to number-letter pairs in the bottom two quadrants, and; (3) to respond to the clockwise rotation. The practice trial restarts if a limit of correct answers is not met. The restart limits are 5 (of 8) correct answers in the first two practice trials and 11 (of 16) correct answers in the third trial. The real task score is the sum of correct responses and could reach a maximum of 65 points.

2.6.1.9. Updating. Updating measures the ability to update information in working memory. We used the updating task adapted from Carretti, Cornoldi, and Pelegrina (2007). The task consists of seven lists of 10 two-digit numbers. For each list the participants are instructed to recall the three smallest numbers in the order of presentation. Before the first number, the symbol "##" is displayed to indicate where the numbers are going to be presented. The numbers are presented sequentially in the middle of the computer screen and range from 15 to 99. The arithmetic distance between within-list numbers has been found by Carretti et al. to affect performance. Therefore the highest and the lowest number within a list just vary between 30 and 36 steps in between and two adjacent numbers vary between two and six steps in between. The three smallest numbers recalled and placed in the correct order of each list were scored with one point and the maximum score for one block with seven lists could reach 21 points.

2.6.2. Self ratings of fatigue

Perceived tiredness and motivation were measured with the Swedish Occupational Fatigue Inventory (SOFI) developed by Åhsberg, Gamberale, and Kjellberg (1995). The questions are based on a factor analysis where five components were established for 25 items describing feelings of cognitive- and physical fatigue. Only the three components describing cognitive fatigue were selected for this study. *Lack of energy* was measured with the words "worn out" and "exhausted", *Lack of motivation* with the words "passive" and "uninterested", and *Tiredness* with the words "sleepy" and "the number of yawns for the last 10 min" on a scale ranged from 1 = not at all to 4 = a lot.

2.6.3. Stress hormones

2.6.3.1. Urinary catecholamines. Urine samples were collected directly when all participants had arrived at the laboratory and at the end of the experimental session. For each participant, the exact time after the bladder had been emptied by voluntary voiding was noted, the total urine volume was measured, and 40 ml of urine was mixed in 100 ml glass bottles containing 0.5 ml 2M HCL as a preservative. Samples were stored in a freezer at −20 °C until subsequent preparation. The samples were thawed within 2 months and purified according to the BIO-RAD method (Urinary Catecholamines by HPLC, Reagent Kit, Catalog Number 195-5841/N) and assayed following standard methods for high performance liquid chromatography. The amounts of catecholamines were expressed as ng/h/kg body weight.

2.6.3.2. Salivary cortisol. Saliva samples were obtained with Salivette tubes (Sarstedt, Landskrona, Sweden) at five times during the experimental session (see Fig. 2). It is unclear when exactly to measure a cortisol response, since the peak cortisol response occurs within different time points (Dickerson & Kemeny, 2004) and there might be a time lag before a decline (as after a restoration period) is measurable. The cortisol samples in our study were taken directly when the participants were seated in the simulated office, after 15 min relaxation (before the work session started), after 2 h of work (before the restoration period), after 7 min restoration, and approximately 10 min after the restoration period ended. The radioimmunoassays (RIA) of the samples were carried out in the Stress Research Institute at Stockholm University by a professional blind to the experimental conditions.

2.7. Procedure

Data collection took place at a simulated office at the University of Gävle. Participants went through a practice session for 1 h, several days in advance of the first experimental session. The practice session was meant to reduce possible training effects on the cognitive tasks. To reduce the risk of variations in outcomes according to the weekly calendar (Larsen & Kasimatis, 1990), the experimental sessions only took place on Tuesdays and Thursdays. The experimental sessions were run between 2.00 and 5.00 pm. For each participant, one day separated the two sessions.

The procedure for an experimental session is presented in Fig. 2. The participants started and ended the session with a urine sample and saliva samples were collected at several time points. This enabled checks on the stress hormone levels at the beginning, middle (after 2 h of work), and end of each session.

To reduce possible high levels of stress caused by earlier activities during the day, the participants were given time to relax at their work station in the laboratory office, after the first urine and saliva sample. During this relaxation period they listened (with headphones) to a story about a fictive culture for 15 min. After the relaxation period they provided a second saliva sample and

completed the SOFI. Before they completed the SOFI, the presentation of the office noise was initiated, and the sound level increased gradually to full strength after 1 min. After the SOFI the participants proceeded through the 2 h of cognitive tasks while one of the two different noise conditions (low or high noise) was played back. Filler tasks were included at different time points to assure that all participants always had something to work with and to make it possible to assemble the participants for the next phase of measurements. Performance on these filler tasks was, however, not further analyzed. The session order of the noise conditions was counterbalanced but not the cognitive tasks within the experimental sessions. The cognitive tasks were in the same order in the two experimental sessions since they were to be performed about the same time in the two noise conditions. The session with cognitive tasks started and ended with the PI and SART tests.

After the cognitive work, a saliva sample was taken and the participants again filled out the SOFI. Next, they put their headphones on and went through the 7-min restoration period with one of the four restorative conditions. During the restoration period the background office noise was canceled. For each of the experimental sessions, the participants were randomly assigned to the restorative conditions (i.e., of eight participants at the eight workstations, two watched the movie of the flowing river with sound, two listened only to the sound of the flowing river, two sat in silence, and two listened to office noise delivered through the headphones). The end of the restoration period was indicated by three beeps in the headphones.

After the restoration period the participants took off their headphones and went through the last dependent measures in silence. The whole procedure took about 3 h to complete, and seven to eight participants were tested at each occasion.

2.8. Statistical analyses

SPSS 18.0 was used for statistical analyses. Nine participants were excluded from analysis because of failures to follow task instructions or equipment malfunctioning, leaving 38 participants for further analysis. Data are missing for the first SOFI measure for an additional 14 participants because of problems with the computer script. One additional value is missing in the SOFI measure of tiredness (yawning). Fourteen values for dopamine and one for norepinephrine were lost to analyses because of procedural failures. Since no effects were found on dopamine when the three urine sample measures were analyzed together, we excluded dopamine in a next step of analysis to regain power. A General Linear Model (SPSS version 18) with repeated measures (mixed) design was used for the analysis of noise and fatigue effects. When Mauchly's test indicated non-sphericity in the variance-covariance matrix for the within-subject analyses, the Greenhouse-Geisser adjusted degrees of freedom and corresponding *p*-values are reported for the *F*-tests.

3. Results

Originally a within-person design was intended for the noise manipulation. However, some of the cognitive measures interacted with the sequence of the noise conditions[1]. In general, the interactions were such that performance in the low noise condition was better when this condition was presented first. This might reflect loss of motivation across sessions. The design employed gives an opportunity to safeguard against these order effects by making a between-subjects design of the performance in the first session only. Accordingly, we treated the noise condition in the first session as a between-subjects manipulation.

3.1. Noise effects on cognitive performance

The between-subjects comparison revealed, in accordance with hypothesis 1, a significant main effect of noise on memory for words (PI-points), $F(1, 30) = 4.27$, $p < .05$, *partial* $\eta^2 = 0.13$, indicating a better performance in low noise compared with high noise (see Fig. 3). The analysis did not reveal an effect on the ability to inhibit information (PI-pro), since few participants wrote words from earlier lists ($F < 1$), but it did reveal a significant effect of noise on the number of missing answers (PI-no answers), $F(1, 30) = 6.62$, $p < .05$, *partial* $\eta^2 = 0.18$. The participants had more no answers (i.e. words not remembered) in high noise compared to low noise. There were, however, no statistically significant effects of noise on the other cognitive measures (all $Fs < 1$).

3.2. Noise effects on acute psychophysiological stress

There were no interactions between noise and session order for the physiological measures. Consequently, noise was treated as a within-subjects factor.

The analysis revealed an effect of time on the cortisol levels $F(1.16, 34.91) = 31.12$, $p < .001$, *partial* $\eta^2 = 0.51$ (see Fig. 4). The participants showed high levels of cortisol when they arrived to the experimental session, and the level had declined in both noise conditions after 2 h of work. There was no statistically significant effect of noise on the decline in the level of cortisol ($F < 1$). We thus did not find support for hypothesis 2.

A significant effect of time was also shown on the norepinephrine levels from the urine samples, $F(1, 35) = 30.40$, $p < 0.001$, *partial* $\eta^2 = 0.47$, with a decline from Pretest to Postrest in both noise conditions (see Fig. 5). There were no statistically significant effects of noise on the level of catecholamines ($F < 1$).

3.3. Noise effects on self ratings of fatigue

For the subjective ratings (SOFI-measures), we found no interactions between noise and session order, and were therefore free to analyze the within-subjects effects of noise. Both noise and time

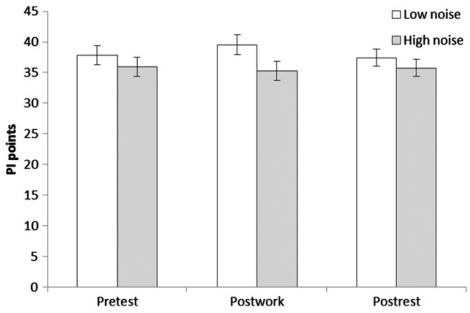


Fig. 3. The number of words remembered at *Pretest, Postwork, and Postrest*in low noise ($n = 18$) and high noise ($n = 20$). Error bars are the standard errors of the mean.

[1] The cognitive measures which showed a significant interaction ($p < 0.05$) with the sequence of the noise conditions presented first: PI-points, $F(1, 36) = 4.49$, partial $\eta^2 = 0.11$; PI-no answers, $F(1, 36) = 6.40$, partial $\eta^2 = 0.15$; Flanker-reaction time, $F(1, 30) = 11.82$, partial $\eta^2 = 0.28$; Logical problems-points, $F(1, 30) = 14.53$, partial $\eta^2 = 0.33$; Serial recall-letter points, $F(1, 30) = 11.04$, partial $\eta^2 = 0.27$; Serial recall-number points, $F(1, 30) = 9.06$, partial $\eta^2 = 0.23$.

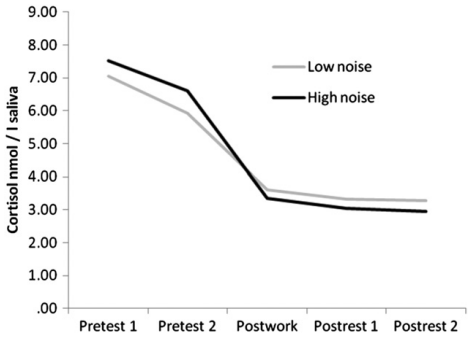


Fig. 4. Cortisol levels (nmol/l saliva) in low- and high noise ($n = 38$) at *Pretest, Postwork, and Postrest.* Note: *Pretest1* = at arriving, *Pretest 2* = after relaxation; *Postrest 1* = after seven minutes of restoration, *Postrest 2* = after additional 5–10 minutes.

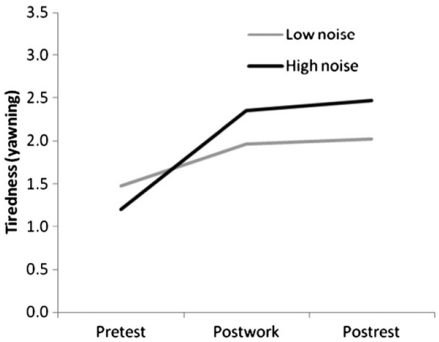


Fig. 6. Self ratings of tiredness (yawning) the last 10 minutes in low- and high noise ($n = 23$) at *Pretest, Postwork, and Postrest.* Note: The scale ranged from 1 = not at all to 4 = a lot.

effects were of interest. In this section we focus on the measures from Pretest (at arrival) to Postwork (after 2 h of work).

The analysis revealed a significant interaction between noise and time on tiredness (yawning), $F(2, 32) = 6.25$, $p < 0.01$, *partial* $\eta^2 = 0.28$; and on lack of motivation (passiveness) $F(2, 32) = 4.33$, $p < 0.05$, *partial* $\eta^2 = 0.21$. In line with hypothesis 3, the participants rated themselves as more tired and less motivated after 2 h of work in high noise compared to low noise (see Figs. 6 and 7, Pretest and Postwork).

3.4. Restoration

When comparing the SOFI scores from Postwork (before the restoration period) to Postrest (after restoration), the analysis revealed a significant between-subjects effect of restorative conditions on change in the ratings of lack of energy (exhaustion), $F(3, 29) = 3.10$, $p < 0.05$, *partial* $\eta^2 = 0.24$; and of lack of motivation (uninterested) $F(3, 29) = 3.02$, $p < 0.05$, *partial* $\eta^2 = 0.24$. Decomposition of this effect using simple effects analysis (LSD) demonstrated that the participants who saw the movie (including river sounds) rated themselves as *having more energy* (i.e., were less exhausted) in comparison with the participants listening to noise ($p < 0.01$) and those listening to river sounds ($p < 0.05$). The participants who listened to noise during the restoration period rated themselves as being *less motivated* (i.e., more uninterested) in comparison with the participants who listened to river sounds ($p < 0.05$) or saw the movie ($p < 0.01$), see Figs. 8 and 9. The post hoc analysis also demonstrated a marginal difference in *tiredness* (yawning) between the movie condition and the noise condition ($p = 0.055$) and between the movie and river sound conditions ($p = 0.052$). An inspection of Table 2 indicates that the participants who heard river sounds or office noise during the restoration period rated their overall experience more negatively than the participants in the other two restoration conditions.

No statistically significant effects from the restoration period were detected on the cognitive or physiological measures.

4. Discussion

In this experimental study we approached two main issues concerning open-plan offices. First, we examined the effects of two levels of office noise on basic working memory processes, self-ratings of mood and fatigue, and physiological measures of stress (salivary cortisol and urinary catecholamines) in a simulated open-plan office. Second, we examined the effects of varying sound conditions on restoration during short period after 2 h of work.

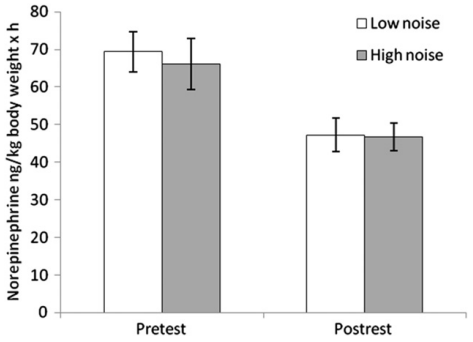


Fig. 5. Norepinephrine levels (ng/kg body weight × height) in low- and high noise ($n = 37$) at *Pretest and Postrest.*

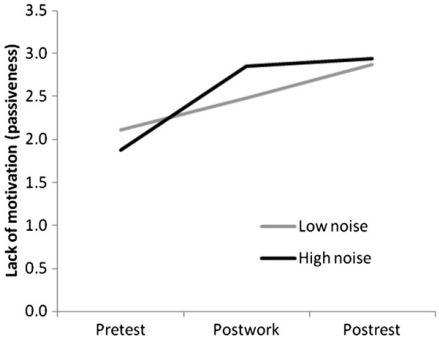


Fig. 7. Self ratings of low motivation (passiveness) in low- and high noise ($n = 24$) at *Pretest, Postwork, and Postrest.* Note: The scale ranged from 1 = not at all to 4 = a lot.

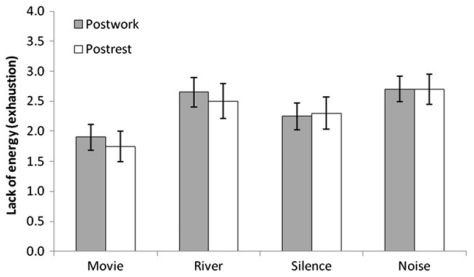


Fig. 8. Self-ratings of lack of energy (exhaustion) before and after 7 minutes of rest in each restorative condition (movie, $n = 10$; river, $n = 8$; silence $n = 9$; and noise, $n = 10$). Error bars are the standard errors of the mean for group comparisons. Note: The scale ranged from 1 = not at all to 4 = a lot.

4.1. Effects of noise on cognitive performance

We employed different cognitive tasks with focus on the ones which have been developed to tap three basic cognitive functions (inhibition, updating and shifting), to see whether some or all of these functions were affected by office noise. We also included other tasks which have been used in earlier studies of office noise such as reading comprehension, serial recall and operation span.

In accordance with Hypothesis 1, there was a main effect of noise on memory for words (PI-points), indicating that performance declined more in high noise compared to low noise. However, no effects were shown on the number of proactive interferences (inhibition), since few participants wrote words from earlier lists. We had not included in the instruction that they always should write five words even when they were not sure or could not remember. There was instead a significant effect of noise on the number of missing answers. However, the number of words the participants did or did not remember cannot tell which specific cognitive processes were involved and therefore affected by noise.

Further, it was unexpected that no other noise effects would be revealed on the additional cognitive measures. This might be explained by limitations of the present study. First, Hongisto (2005) has suggested that it is not the noise level of office noise that affects cognitive performance but the intelligibility of speech, or STI if we want to put it into a physical measure. The STI of the low noise condition did not differ much from the high noise condition although the overall noise level was reduced by 12 dB (Table 1). According to Hongisto, the cognitive performance improves only if STI reduces from highly intelligible speech (STI at least 0.60) to poorly intelligible speech (STI less than 0.30). This may be one of

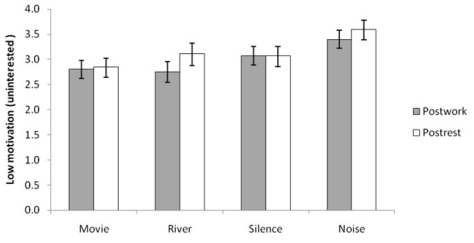


Fig. 9. Self-ratings of low motivation (uninterested) before and after 7 minutes of rest in each restorative condition (movie, $n = 10$; river, $n = 8$; silence $n = 9$; and noise, $n = 10$). Error bars are the standard errors of the mean for group comparisons. Note: The scale ranged from 1 = not at all to 4 = a lot.

Table 2
The mean change in the SOFI-measures in different restorative conditions. Negative values indicate decrease from before to after the rest period (i.e. improvement).

	Movie	River	Silence	Noise
Sleepy	−.40	.00	−.44	−.45
Exhaustion	−.20	−.13	.11	.18
Passive	.00	−.25	.11	.27
Yawning	.10	.25	−.56	−.18
Worn out	−.10	.25	−.11	.00
Uninterested	.30	.25	−.22	.27
Mean	−.05	.06	−.19	.02

Note: The scales ranged from 1 = not at all to 4 = a lot.

the reasons why some of the cognitive measures were independent of our noise manipulation. Second, the exposure times can be critical for performance effects to appear, since research has shown that it is possible to compensate for adverse sound conditions during shorter exposures by exerting extra effort (Kjellberg, 1997). The participants also knew the duration of the session which might have mitigated their tolerance of the noise exposure. In the long run, however, consequences like fatigue and less efficiency might emerge and affect performance. This could have been the case if the exposure time in this study had been longer than 2 h, since the subjective ratings indicated fatigue effects. Interesting results of this study is that noise level and absorption proved to be of importance for memory performance. These results, however, are different from some earlier findings. Ellermeier and Hellbrück (1998) found for example no effects of office noise level (75 dBA and 60 dBA overall level) on cognitive performance. Further research is needed to further explore the relation between cognitive performance and varying acoustic conditions (level, STI, fluctuation etc.) in order to find a good balance for performance in open-plan offices.

4.2. Effects of noise on acute stress effects

The participants showed a significant decline in cortisol in both noise conditions over time, consistent with the decline that occurs as a part of normal circadian variation. The same pattern is shown in norepinephrine measured from urine. The results thus give no support for Hypothesis 2. Further studies need to address the question on how to deal with high stress levels from start. According to this study 15 min relaxation in the beginning was not enough.

4.3. Effects of noise on self ratings of fatigue

In line with Hypothesis 3, the participants felt more tired and less motivated after they had worked for 2 h in high noise compared to low noise. This is in line with other studies indicating motivational deficits, irritability and fatigue among groups exposed to noise (reviewed by Kjellberg, 1997).

Chigot (2005) has in a review indicated that subjective experience seldom paralleled the objective assessment of the effects of noise in offices, and this seems to have been the case also in this study. Further research should therefore continue to explore the effects of noise with different measures.

4.4. Restoration

After 2 h of work the participants had 7 min to restore. The participants who saw a movie of a flowing river with corresponding sound reported having more energy compared to the participants who just listened to river sounds, or those who continued to listen to office noise. The participants who listened to office noise during

the restoration phase also experienced themselves to be less motivated after this period, compared to the participants who listened to river sounds, and those who saw the nature movie with river sounds. These results support Hypothesis 4 by indicating that the sound conditions may promote different restorative experiences. There were, however, no significant effects of restoration condition on the cognitive or physiological measures, which might be due to the small sample for comparison in each restorative condition and/or the brief period allowed for restoration. Increasing the restorative period would be a good starting point for clarifying within which timespan different effects start to emerge (cf. Hartig et al., 2003). However, the timespan needed probably depends on the antecedent condition of fatigue and the environmental conditions investigated.

To sum up, further research needs to address the issues of which cognitive processes get fatigued with noise, how long a restorative phase is needed to get the specified processes restored, and which environmental sounds promote restoration.

4.5. Conclusion and practical implications

The design of work environments has significance for employee wellbeing and performance. In this study the results showed that the participants' memory performance declined more in high noise compared to low noise, and that they experienced more tiredness and less motivation with high noise. Significant reduction of noise level can reduce the negative effects of background noise in open-plan offices. This study included a reduction of noise level by 12 dB which is very challenging to reach in office environments. It requires simultaneous consideration of many noise control measures (Virjonen, Keränen, & Hongisto, 2009) and silent behavior in the office (office etiquette).

We saw no reliable noise effects on the physiological measures. This finding suggests that the effects of short-term noise exposure on cognitive performance are not mediated by a change in physiology. In real-life long-term chronic noise exposure this may be different but to what extent we cannot say from the present study.

Further research is needed to find the best combination of reduced noise level, speech intelligibility and intermittent sounds to enhance performance and well-being in open-plan offices. There should also be further concern for designing sound environments that can promote daily restoration at workplaces. Rest in office noise affected motivation in a negative way. This study also indicated that a movie of a flowing river with corresponding water sound is experienced as giving more energy than staying in continued office noise or just listening to river sounds for 7 min.

So far, many environments have just been studied according to their visual value for restoration. Therefore, multi-modal restoration also needs to be addressed in a larger context than at work. Looking at a combination of senses might indicate that a nature environment such as a noisy park might restrict restoration while a quiet park environment, where only nature sounds are heard, might promote restoration. When comparing different environments, it might also turn out that a calm café with soft music in a noisy urban environment might be more restorative than going to a crowded and noisy park, which would contrast with earlier results. The noise level increments at workplaces and in urban areas demand compensatory strategies that include access to restorative environments. These are health issues that need to be further addressed.

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References

Åhsberg, E., Gamberale, F., & Kjellberg, A. (1995). *Upplevd trötthetskvalitet vid olika arbetsuppgifter. Utveckling av ett mätinstrument.* [*Perceived fatigue quality during different work tasks. Development of a measure*]. Solna: Arbetsmiljöinstitutet. nr. 20.

Baddeley, A. D. (2000). The episodic buffer: A new component of working memory? *Trends in Cognitive Sciences, 4*, 417–423.

Baddeley, A. D. (2002). Is working memory still working? *European Psychologist, 7*, 85–97.

Baddeley, A. D., & Hitch, G. J. (1974). Working memory. In G. A. Bower (Ed.), *The Psychology of learning and motivation* (pp. 47–89). New York, N.Y: Academic Press.

Banbury, S., & Berry, D. C. (1998). Disruption of office-related tasks by speech and office noise. *British Journal of Psychology, 89*, 499–517.

Banbury, S. P., Macken, W. J., Tremblay, S., & Jones, D. M. (2001). Auditory distraction and short-term memory: Phenomena and practical implications. *Human Factors: The Journal of the Human Factors and Ergonomics Society, 43*, 12–29.

Berman, M. G., Jonides, J., & Kaplan, S. (2008). The cognitive benefits of interacting with nature. *Psychological Science, 19*, 1207–1212.

Berto, R. (2005). Exposure to restorative environments helps restore attentional capacity. *Journal of Environmental Psychology, 25*, 249–259.

Carretti, B., Cornoldi, C., & Pelegrina, S. L. (2007). Which factors influence number updating in working memory? The effects of size distance and suppression. *The British Psychological Society, 98*, 45–60.

Chigot, P. (2005). Effects of sound in offices: subjective experience vs. objective assessment. *Facilities, 23*, 152–163.

Daneman, M., & Carpenter, P. A. (1980). Individual differences in working memory and reading. *Journal of Verbal Learning and Verbal Behavior, 19*, 450–466.

De Croon, E. M., Sluiter, J. K., Kuijer, P. P., & Frings-Dresen, M. H. W. (2005). The effect of office concepts on worker health and performance: A systematic review of the literature. *Ergonomics, 48*, 119–134.

Dickerson, S. S., & Kemeny, M. E. (2004). Acute stressors and cortisol responses: A theoretical integration and synthesis of laboratory research. *Psychological Bulletin, 130*, 355–391.

Ellermeier, W., & Hellbrück, J. (1998). Is level irrelevant in "irrelevant speech"? Effects of loudness, signal-to-noise ratio, and binaural unmasking. *Journal of Experimental Psychology, 24*, 1406–1414.

Eriksen, B. A., & Eriksen, C. W. (1974). Effects of noise letters upon the identification of a target letter in a nonsearch task. *Perception & Psychophysics, 16*, 143–149.

Evans, G. W., & Johnson, D. (2000). Stress and open-office noise. *Journal of Applied Psychology, 85*, 779–783.

Friedman, N. P., & Miyake, A. (2004). The relations among inhibition and interference control functions: A latent-variable analysis. *Journal of Experimental Psychology, 133*, 101–135.

Haka, M., Haapakangas, A., Keränen, J., Hakala, J., Keskinen, E., & Hongisto, V. (2009). Performance effects and subjective disturbance of speech in acoustically different office types – A laboratory experiment. *Indoor Air, 19*, 454–467.

Hartig, T., Böök, A., Garvill, J., Olsson, T., & Gärling, T. (1996). Environmental influences on psychological restoration. *Scandinavian Journal of Psychology, 37*, 378–393.

Hartig, T., Evans, G. W., Jamner, L. D., Davis, D. S., & Gärling, T. (2003). Tracking restoration in natural and urban field settings. *Journal of Environmental Psychology, 23*, 109–123.

Hartig, T., Mang, M., & Evans, G. W. (1991). Restorative effects of natural environment experiences. *Environment and Behavior, 23*, 3–26.

Hedge, A. (1982). The open-plan office: A systematic investigation of employee reactions to their work environment. *Environment and Behavior, 14*, 519–542.

Heitz, R. P., & Engle, R. W. (2007). Focusing the spotlight: Individual differences in visual attention control. *Journal of Experimental Psychology, 136*, 217–240.

Hongisto, V. (2005). A model predicting the effect of speech of varying intelligibility on work performance. *Indoor air, 15*, 458–468.

Jones, D. (1990). Recent advances in the study of human performance in noise. *Environment International, 16*, 447–458.

Jones, D., & Morris, N. (1992). Irrelevant speech and serial recall: Implications for theories of attention and working memory. *Scandinavian Journal of Psychology, 33*, 212–229.

Just, M. A., & Carpenter, P. A. (1992). A capacity theory of comprehension: Individual differences in working memory. *Psychological Review, 99*, 122–149.
Khalfa, S., Bella, S. D., Roy, M., Peretz, I., & Lupien, S. J. (2003). Effects of relaxing music on salivary cortisol level after psychological stress. *Annals New York Academy of Science, 999*, 374–376.
Kjellberg, A. (1997). Noise. In H. A. Waldron, & C. Edling (Eds.), *Occupational health practice* (4th edition). (pp. 241–256). Oxford: Butterworth Heinemann.
Kjellberg, A., & Sköldström, B. (1991). Noise annoyance during the performance of different nonauditory tasks. *Perceptual and Motor Skills, 73*, 39–49.
Kristiansen, J., Mathiesen, L., Nielsen, P. K., Hansen, A. M., Shibuya, H., Petersen, H. M., et al. (2008). Stress reactions to cognitively demanding tasks and open-plan office noise. *International Archives of Occupational and Environmental Health, 82*, 631–641.
Larsen, R. J., & Kasimatis, M. (1990). Individual differences in entrainment of mood to the weekly calendar. *Journal of Personality and Social Psychology, 58*, 164–171.
Laumann, K., Gärling, T., & Stormark, K. M. (2001). Rating scale measures of restorative components of environments. *Journal of Environmental Psychology, 21*, 31–44.
Laumann, K., Gärling, T., & Stormark, K. M. (2003). Selective attention and heart rate responses to natural and urban environments. *Journal of Environmental Psychology, 23*, 125–134.
Lustig, C., May, C. P., & Hasher, L. (2001). Working memory span and the role of proactive interference. *Journal of Experimental Psychology, 103*, 199–207.
Manly, T., Robertson, I. H., Galloway, M., & Hawkins, K. (1999). The absent mind: further investigations of sustained attention to response. *Neuropsychologia, 37*, 661–670.
Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex "frontal lobe" tasks: A latent variable analysis. *Cognitive Psychology, 41*, 49–100.
Navai, M., & Veitch, J. A. (2003). *Acoustic satisfaction in open-plan offices: Review and recommendations (IRC-RR-151)*. National research council Canada: Institute for Research in Construction.
Oommen, V. G. O., Knowles, M. K., & Zhao, I. Z. (2008). Should health service managers embrace open plan work environments? A review. *Asia Pacific Journal of Health Management, 3*, 37–43.
Peterson, L., & Peterson, M. J. (1959). Short-term retention of individual verbal items. *Journal of Experimental Psychology, 58*, 193–198.
Rashid, M., & Zimring, C. (2008). A review of the empirical literature on the relationships between indoor environment and stress in health care and office settings: Problems and prospects of sharing evidence. *Environment and Behavior, 40*, 151–190.
Rogers, R. D., & Monsell, S. (1995). Costs of a predictable switch between simple cognitive tasks. *Journal of Experimental Psychology, 124*, 207–231.
Roper, K. O., & Juneja, P. (2008). Distractions in the workplace revisited. *Journal of Facilities Management, 6*, 91–109.
Schlittmeier, S., Hellbrück, J., Thaden, R., & Vorländer, M. (2008). The impact of background speech varying in intelligibility: Effects on cognitive performance and perceived disturbance. *Ergonomics, 51*, 719–736.
Sloan, R. P. (1991). Cardiovascular effects of noise. In T. H. Fay (Ed.), *Noise and health* (pp. 15–26). New York: The New York Academy of Medicine.
Smith-Jackson, T. L., & Klein, K. W. (2009). Open-plan offices: Task performance and mental workload. *Journal of Environmental Psychology, 29*, 279–289.
Sörqvist, P., Halin, N., & Hygge, S. (2009). Individual differences in susceptibility to the effects of speech on reading comprehension. *Applied Cognitive Psychology, 24*, 67–76.
Turner, M. L., & Engle, R. W. (1989). Is working memory capacity task dependent? *Journal of Memory & Language, 28*, 127–154.
Ulrich, R. S., Simons, R. F., Losito, B. D., Fiorito, E., Miles, M. A., & Zelson, M. (1991). Stress recovery during exposure to natural and urban environments. *Journal of Environmental Psychology, 11*, 201–230.
Venetjoki, N., Kaarlela-Tuomaala, A., Keskinen, E., & Hongisto, V. (2006). The effect of speech and speech intelligibility on task performance. *Ergonomics, 49*, 1068–1091.
Virjonen, P., Keränen, J., & Hongisto, V. (2009). Determination of acoustical conditions in open-plan offices – Proposal for new measurement method and target values. *Acta Acustica united with Acustica, 95*, 279–290.
Waye, K. P., Bengtsson, J., Rylander, R., Hucklebridge, F., Evans, P., & Clow, A. (2002). Low frequency noise enhances cortisol among noise sensitive subjects during work performance. *Life Sciences, 70*, 745–758.